BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Debugging Syntax and Logic Errors

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. For valid ages 13-18, identify the two valid boundary values.

2. What is an IDE's editor used for?
   A) Writing source code
   B) Fixing physical CPU faults
   C) Managing user access rights
   D) Defragmenting a hard drive

3. "The first index in every OCR ERL array is 1." Correct this. Explain why.

4. A ticket program checks ages. "A few successful tests prove it has no faults." Correct this. How should it be tested?

Today's objectives

Explain the core ideas of debugging syntax and logic errors using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Syntax error
Code breaks language grammar and may fail to run.

Logic error
Code runs but behaves incorrectly.

Trace and isolate
Follow values and conditions to locate the first incorrect change.

Repair and retest
Correct the cause, then check both the original and adjacent boundary cases.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Syntax error
Code breaks language grammar and may fail to run.

Logic error
Code runs but behaves incorrectly.

Trace and isolate
Follow values and conditions to locate the first incorrect change.

Repair and retest
Correct the cause, then check both the original and adjacent boundary cases.

I Do - Worked model

A loop uses total = total - value when calculating a sum. Diagnose and correct it.

- The program runs, so it may be a logic error.
- Trace a small input to observe subtraction.
- Replace subtraction with total = total + value.
- Retest several inputs and compare totals with expected sums.

We Do - Guided practice

1. A range check uses OR instead of AND for score >=0 and score <=100. Fix it.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the Boolean operator that makes the range test accept every number.

```
octet = 300
if octet >= 0 OR octet <= 255 then
print("Valid")
else
print("Invalid")
endif
```

You Do - Independent exam questions

Q2. Differentiate a syntax error from a logic error. [2 marks]

Q3. Identify the fault in if age >= 13 OR age <= 18 for acceptance age 13-18. [2 marks]

Q4. Explain why testing after debugging is necessary. [2 marks]

Exit Ticket - True or False?

A logic error can occur in a program that runs without a syntax error.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. 13 and 18 lie on the edges of the valid range.

2. A - Writing source code

3. A fixed-size indexed collection of related elements; OCR ERL indexing begins at zero.

4. Test valid and invalid inputs. Compare actual and expected results to find faults. A few successful tests cannot prove there are no faults.

I Do - Explain the model

The program runs, so it may be a logic error.

Trace a small input to observe subtraction.

Replace subtraction with total = total + value.

Retest several inputs and compare totals with expected sums.

We Do - Model answers

1. Retest -1, 0, 100 and 101.

Code First - Answer

```
octet = 300
if octet >= 0 AND octet <= 255 then
print("Valid")
else
print("Invalid")
endif
```

Expected output: Invalid

You Do - Self-mark

Q2: Syntax violates language rules; logic produces incorrect behaviour even if program runs.

Q3: Use AND, not OR; OR incorrectly accepts values outside the range.

Q4: To verify the correction and ensure no new faults were introduced.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.